

An Open Implementation of the TEI Processing Model

A Call for Institutional Support

1 The Challenge

The TEI Processing Model is an open standard, defined in the TEI Guidelines (Chapter 23.5.4), that describes how TEI-encoded texts should be transformed into output formats such as HTML, PDF, or LaTeX. It was designed to free edition projects from writing thousands of lines of custom XSLT – instead, transformation rules are declared in ODD files using TEI’s own vocabulary.

Despite being an open standard since 2016, there is currently **only one complete implementation**: TEI Publisher, built on the XML database eXist-db. This means that anyone who wants to use the Processing Model must operate eXist-db – a specialised infrastructure that many projects, especially smaller ones, cannot afford to maintain.

At the same time, **Python has become the lingua franca of the Digital Humanities**. It is the language in which most DH researchers are trained, in which corpus linguistics, NLP, and machine learning take place, and in which tools like CollateX, spaCy, NLTK, and Jupyter Notebooks operate. Yet there is no way to use the TEI Processing Model in these environments.

A standard with only one implementation is, in practice, a proprietary format.

2 The Project

e-editiones – the Swiss non-profit behind TEI Publisher – proposes to create:

The project is structured as two self-contained, independently fundable sub-projects. Each delivers a usable output on its own; together they form the complete intervention. Funding can start from a minimum of CHF 30,000 for Sub-project 1 and extend modularly.

1. **Sub-project 1 – A Python implementation of the TEI Processing Model** – a lightweight, open-source library («pip install») that transforms TEI documents according to ODD specifications, without requiring eXist-db. Outputs: HTML, LaTeX, plain text. Usable as a library, CLI tool, or in existing workflows (Jupyter, CI/CD pipelines, serverless deployments). *Minimum: CHF 30,000; Full: CHF 42,000.*
2. **Sub-project 2 – An abstraction layer in `tei-publisher-lib`** – decoupling the existing XQuery core library from eXist-db-specific APIs, enabling alternative XQuery engines (BaseX, Saxon) in the future. *Minimum: CHF 24,000; Full: CHF 32,400.*
3. **Direct collaboration with the TEI Technical Council** – contributing into the Council's ongoing work on the next iteration of the Processing Model specification, based on findings from the second, independent implementation: ambiguities, gaps, and de-facto extensions are surfaced and brought back as concrete proposals. Two Steering Committee members are current or former Council members. Included proportionally in both sub-projects.

2.1 Project Team

Project leads: Wolfgang Meier (creator of TEI Publisher) and Magdalena Turska (TEI Processing Model co-author, former TEI Technical Council member).

Steering Committee: Dr. Helena Bermúdez Sabel (JinnTec, TEI Technical Council), Dr. Anne Diekjobst (Co-President e-editiones, ULB Bonn), Dr. Andreas Kränzle (Co-President e-editiones, Karl Barth Edition), Dr. Andreas Wagner (Max-Planck-Institut für Rechtsgeschichte und Rechtslehre, Frankfurt).

2.2 At a Glance

Structure:	Two self-contained, independently fundable sub-projects
Entry budget:	From CHF 30,000 (Sub-project 1, Minimum)
Full build-out:	Up to CHF 75,000 (both sub-projects, full)
Sponsor:	e-editiones (Swiss non-profit association)
Outputs:	Open-source Python library on PyPI, abstraction layer in <code>tei-publisher-lib</code> , contributions to the TEI Technical Council on the Processing Model specification

3 Why This Matters

A standard needs more than one implementation.

A second, independent implementation forces clarity: where is the spec precise, where is it ambiguous, where do TEI Publisher extensions fill gaps that should be standardised? The TEI Technical Council can only improve the specification if such questions are raised systematically.

It lowers the barrier to entry.

Instead of deploying eXist-db, a researcher can «pip install» and process their TEI documents in a Jupyter Notebook. For smaller edition projects, dissertation-level work, and teaching contexts, this is transformative.

It meets the community where they work.

Python-based DH workflows are the daily reality for most researchers. Making the Processing Model available in Python will significantly increase its adoption – and with it, the adoption of TEI-based best practices for encoding cultural heritage.

It strengthens TEI Publisher, not weakens it.

Projects that start with the lightweight Python tool can move to TEI Publisher when they need a full publishing platform – because their ODDs are compatible. A broader ecosystem benefits everyone.

It secures cultural heritage infrastructure.

Scholarly editions are long-term investments in cultural heritage. They deserve infrastructure that is not dependent on a single platform. Diversity of implementations is resilience.

About e-editiones: e-editiones (<https://e-editiones.org>) is a Swiss non-profit dedicated to the cooperative development of open-source tools for digital scholarly editions. It coordinates the development of TEI Publisher – the de facto standard for TEI-based publishing in Switzerland – and acts as project sponsor. The e-editiones project registry lists over 40 edition projects worldwide that use TEI Publisher (<https://www.e-editiones.org/map/>).

4 How You Can Support This Project

We are seeking support from institutions that share our commitment to open standards and sustainable digital infrastructure for the humanities.

4.1 Financial Contribution

We invite institutional partners to co-fund the project. Contributions of any size are welcome.

Partnership Levels		
Level	Contribution	Recognition
Sustaining Partner	CHF 10,000+	Logo on landing page, acknowledgment in all publications, named in the PyPI package metadata, seat on the Steering Committee
Supporting Partner	CHF 5,000–9,999	Logo on landing page, acknowledgment in all publications
Contributing Partner	CHF 1,000–4,999	Name on landing page, acknowledgment in final report
Friend of the Project	up to CHF 999	Name on landing page
All financial contributions are administered by e-editiones, a tax-exempt non-profit association under Swiss law. Contributions may be tax-deductible depending on your jurisdiction.		

4.2 Letter of Support

If your institution cannot contribute financially but values this work, a **Letter of Support** strengthens our applications to foundations. A letter should briefly state your institution's relationship with TEI, why a second Processing Model implementation would benefit your work, and your endorsement of the project. Letters can be sent to info@e-editiones.org.

4.3 Become a Member of e-editiones

Becoming a member of e-editiones is a durable form of support. Members sustain the ongoing infrastructure behind TEI Publisher and this project, help shape its strategic direction, and join a growing community of scholarly editing practitioners. Individual and institutional memberships are available. Details and application: e-editiones.org/pages/how-to-become-a-member.

4.4 In-Kind Contributions

You can also support the project by providing **test data** (ODD files and TEI documents as test cases), participating in **beta testing**, or helping with **dissemination** through your networks and conferences.